

Bezier Curve Motion

File path: Assets/UIAnimationToolkit/Editor/Animation/Anim_BezierCurveMotion.cs

Menu: Tools > UI Animation Toolkit > Animation > Bezier Curve Motion

Overview

Edit a multi-segment 2D Bezier path in the Scene view, then bake a UI object’s movement into an AnimationClip. Optional Z orientation follows the path tangent. The current version also includes keyframe reduction after baking.

How to use

1. Open the destination clip in the Animation window. Position its playhead at the desired start time.
2. Assign the target GameObject in Target Object; this workflow is intended for UI objects with a RectTransform.
3. Drag the white anchors and blue control handles in the Scene view to shape the path. Add or remove the last segment as needed.
4. Set Duration (s), Frame Rate (fps) and Easing Curve.
5. Optionally enable Align Z Rotation to Path and choose Rotation Offset.
6. Click Apply to Clip to bake X/Y position and optional Z rotation. Optionally use Reduce Keyframes afterward and inspect the result.

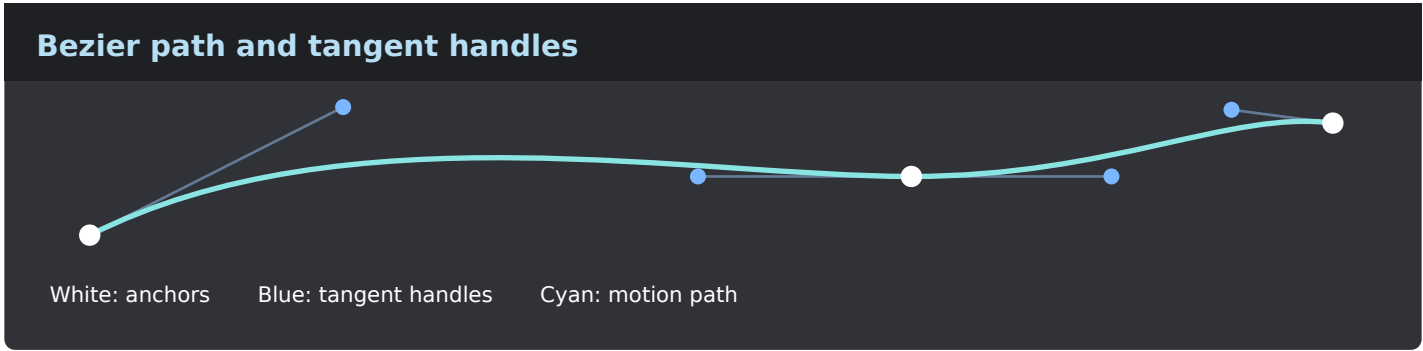
Interface and parameters

Path Anim Tool

Target Object	Animated UI object
Animation Clip (Auto)	Menu_Open.anim
Start Time	Animation-window playhead
Duration / Frame Rate	1 s / 30 fps
Orientation	Align Z Rotation to Path
Path	White anchors / blue handles

Apply to Clip

English interface schematic replacing the supplied mixed-language screenshot.



Schematic: anchors and tangent handles define a multi-segment 2D motion path.

Control / parameter	Description
Target Object	Object whose motion will be baked.
Animation Clip (Auto)	Read-only clip obtained from the Animation window.
Start Time	Uses the Animation-window playhead at the time of baking.
Duration / Frame Rate	Segment duration and sample rate; use a rate appropriate to the destination clip.
Easing Curve	Control how progress travels along the path.
Align Z Rotation to Path	Orient Z rotation to the path tangent.
Rotation Offset	Additional angle applied to the tangent direction.
Control points / segment buttons	Edit positions in the Scene view or fields; add a segment or remove the final segment.
Show Preview	Show the Scene path and direction preview.
Position Tolerance (%)	Error tolerance for position-curve key reduction, relative to value range.
Rotation Tolerance	Angular tolerance for rotation reduction when orientation is enabled.
Reduce Keyframes	Simplify baked curves using RDP and recalculate tangents; inspect the approximation.

Notes

- Path coordinates are authored in Scene world space and converted to the target parent’s local space when written.
- Existing position keys inside the baked time interval can be replaced. Work on a duplicate clip when comparing paths.
- Scene-handle path edits do not implement the same Undo workflow as clip writes; do not rely on Ctrl+Z to restore a dragged path point.
- The tool creates 2D paths and optional Z rotation, not a general 3D path-following system. Larger reduction tolerances can change the motion.